

PPO Learn more from each rollout

PPO-Clip: useful updates with controlled incentives

 Large updates can undo useful behavior

Unconstrained policy gradient



Large shift

Clipped surrogate objective



Repeated updates can overshoot.

Limit incentives for large changes.

Clipping shapes the objective; it does not impose a hard policy-distance bound.

 Make expensive interaction data go further

One update per collected batch



One
pass

Rollouts dominate the data budget.

Several mini-batch epochs



Reuse the batch, then recollect.

Keep old-policy probabilities fixed.



A practical alternative to trust-region solvers



TRPO

Constrained optimization

More involved solver



PPO-Clip

First-order mini-batch updates

Actor-critic + simple clipping

Balance simplicity, sample reuse and stable learning



Still needs fresh rollouts and a useful advantage estimate.